

TIC 232039858

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 6.478900 d
Epoch (t0) 1473.1396 BTJD
Duration 0.0840 d
Depth 29.358 % (293,577 ppm)
SNR 454.9

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

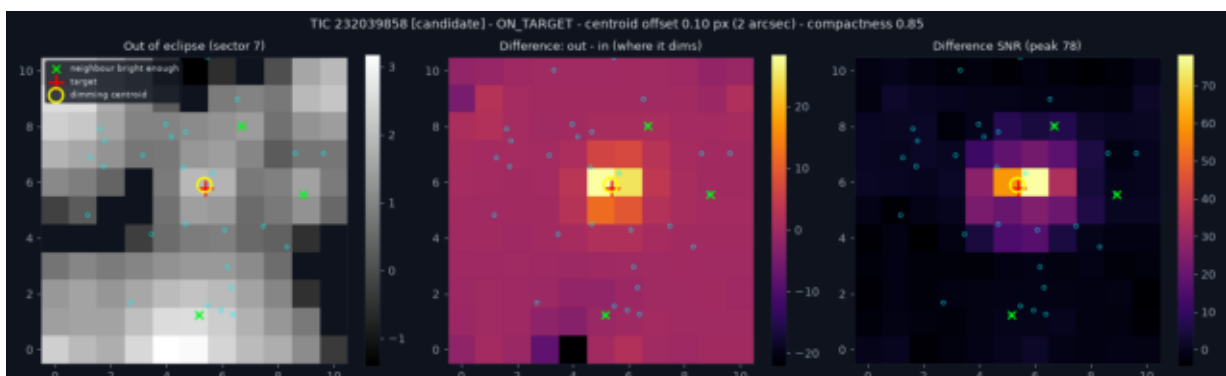
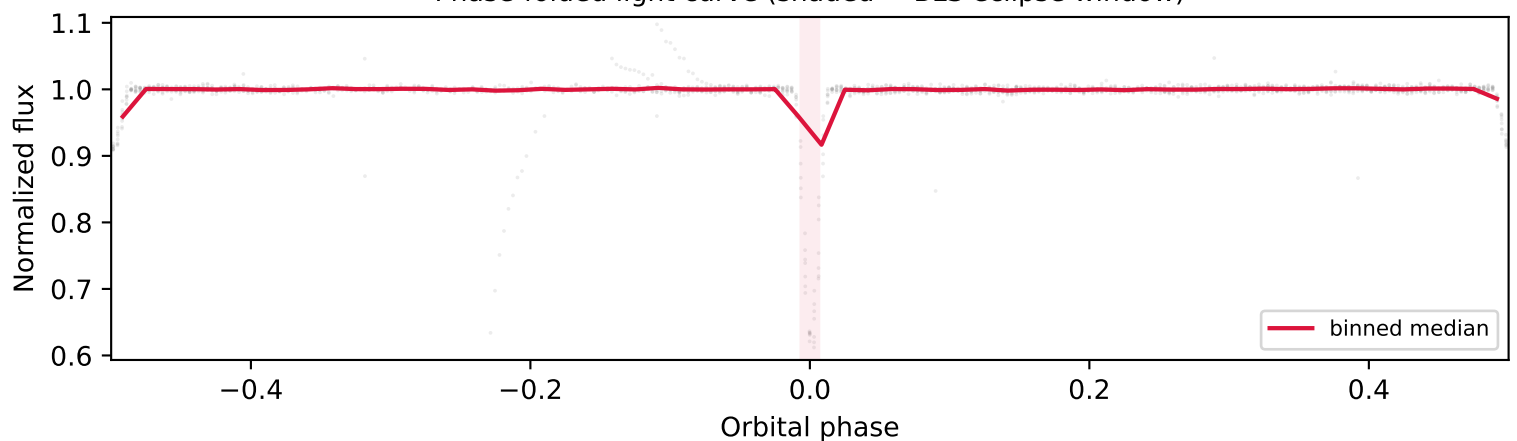
Eclipses observed 7 (gate ≥ 3)
Robust depth ratio 1.02 (gate ≥ 0.50)
Robust depth sigma 351.5 (gate ≥ 7.0)
Odd/even z-score 0.27 (consistent)
Secondary eclipse detected, phase -0.49, 8.2 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 7 light-curve dip in this sector: 303.7 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 78.34 (gate ≥ 5.0) Compactness 0.85 (gate ≥ 0.50)
Dimming centroid offset 0.10 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 232039862
2.50 px away, dTmag +1.30 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.